

# Summary of Changes in z\_score\_regularized\_final.py

## Overview

The original implementation calculated z-scores as (repeat count – population median) / population SD and converted them into one-sided normal p-values. This often produced too many significant tandem repeats because many loci have extremely small population standard deviations. The updated implementation introduces statistically motivated regularization to produce more conservative and biologically meaningful significance estimates.

### 1. Adaptive variance regularization

Instead of using the reported SD directly, the script computes an effective SD:  $\sigma_{\text{eff}} = \sqrt{\text{SD}^2 + \lambda^2}$ , where  $\lambda = \max(0.5, 0.05 \times \text{population median}, 0.25 \times \text{population SD})$ . This reduces inflated z-scores caused by unrealistically small SDs while leaving highly variable loci largely unchanged.

### 2. Minimum SD floor

Population SD is constrained to a minimum of 0.5 repeat units before regularization to prevent numerical instability and excessive z-scores from nearly invariant loci.

### 3. Minimum biological effect size

Differences smaller than one repeat unit from the population median are treated as biologically insignificant. For these loci, the script assigns  $z = 0$  and  $p = 1$  instead of reporting statistical significance.

### 4. Two-sided hypothesis testing

The original script used one-sided p-values that primarily detected repeat expansions. The updated implementation uses two-sided p-values, allowing both repeat expansions and contractions to be detected.

### 5. Improved statistical robustness

These changes reduce false positives, decrease z-score inflation at low-variance loci, preserve ranking of meaningful loci, and improve the biological interpretation of significant tandem repeats.

### 6. Workflow compatibility

No changes are required to the existing Nextflow pipeline. The command line interface, INFO fields, VCF parsing, motif matching, BH correction, and output format remain unchanged.

## Expected impact

The updated method is expected to reduce inflated significance, improve calibration of z-scores and p-values, and prioritize loci showing both statistically and biologically meaningful deviations from the reference population.